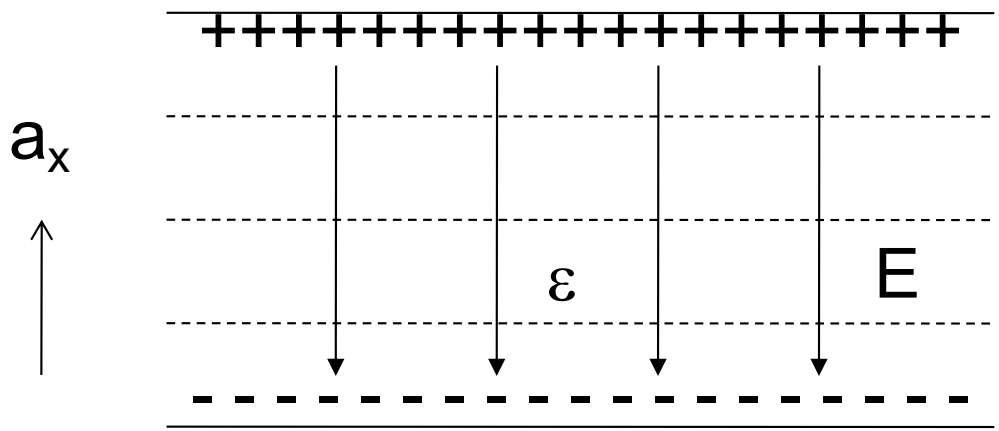
ECE 329
Lectures 10

Capacitance and Conductance

Dr. Victoria Shao
yangshao@Illinois.edu

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Parallel-plate Capacitor



x=d, Q

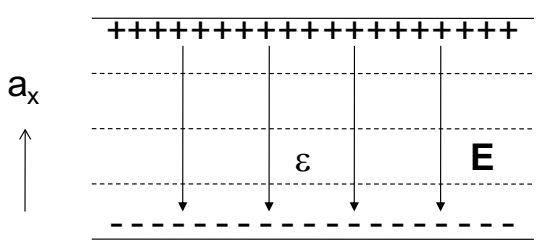
Equipotential lines

x=0, -Q

Capacitance is: $C = Q/V$

Steps to Find Capacitance

Given V, unknown Q



ε is constant and
ρ=0 in between

- Laplace Equation $\nabla^2 V = 0$
- Find V using boundary conditions

$$V(z) = V_0 \frac{x}{d}$$

- Find **E** using $\vec{E} = -\nabla V$
- Find **D** using $\vec{D} = \varepsilon \vec{E}$

$$\vec{E} =$$

$$\vec{D} =$$

- Get surface charge density on one conductor using BC

$$\rho_s = \vec{a}_n \bullet (D_{n1} - D_{n2})$$

$$\rho_s =$$

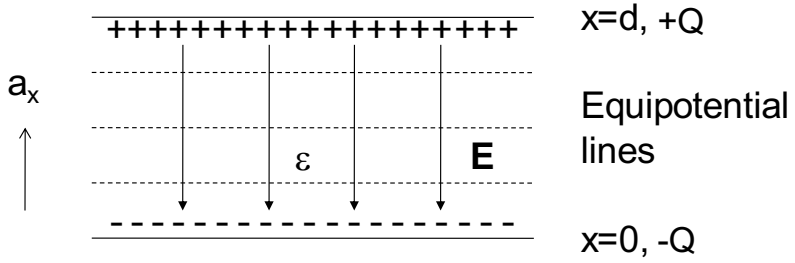
$$Q =$$

- Total charge $Q = (Area)(\rho_s)$
- Capacitance $C = Q/V$

$$C = \frac{\varepsilon A}{d}$$

Steps to Find Capacitance

Given Q, unknown V



ε is constant and ρ=0 in between

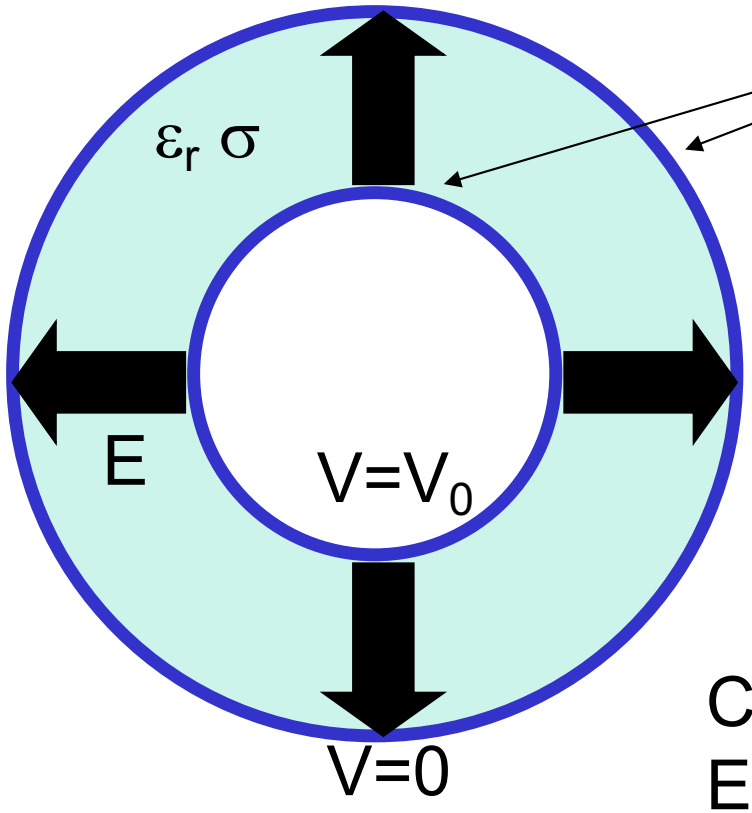
- Find $\vec{D}$ using $\vec{D} = \frac{\rho_s}{2}\,\mathrm{sgn}(x)\,\hat{x}$ → $\vec{D} =$
- Find $\vec{E}$ using $\vec{D} = \epsilon\vec{E}$ → $\vec{E} =$
- Calculate voltage drop → $V =$

$$V = \int_0^d \vec{E} \cdot d\vec{l}$$

- Capacitance

$$C = Q/V \quad \rightarrow \quad C = \frac{\varepsilon A}{d}$$

Coaxial Cable



Perfect conductors

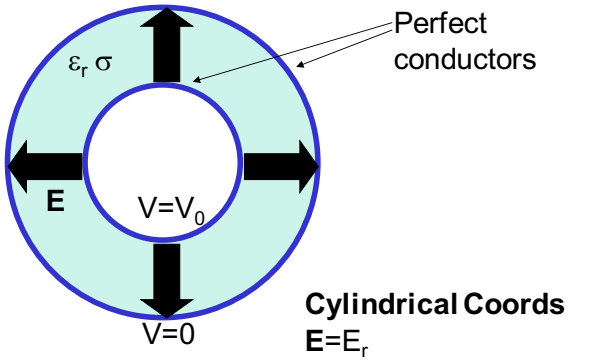
Cylindrical Coordinates

$E=E_r$

z direction along length of the cable

Coaxial Cable

Given V, unknown Q



Cylindrical Coords
E=E_r
z direction along length of the cable

$$\nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0$$

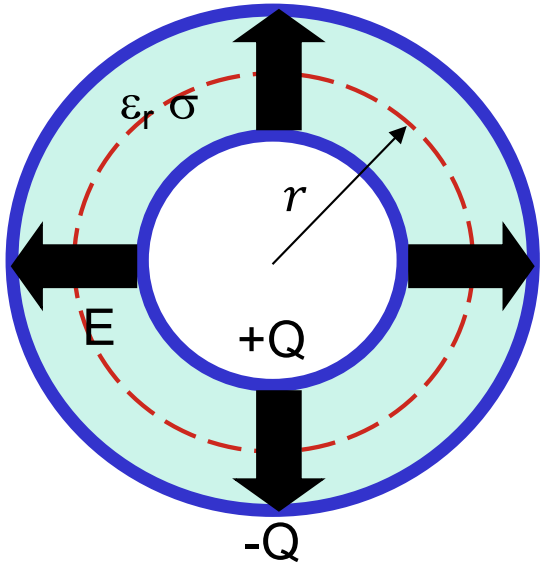
$$\rho = \begin{cases} \varepsilon V_0 /(a\ln(b/a)), r = a \\ -\varepsilon V_0 /(b\ln(b/a)), r = b \end{cases}$$

$$C = \frac{2\pi\varepsilon L}{\ln(b/a)}$$

Coaxial Cable Capacitance

Given Q, unknown V

(Review at home)



Cylindrical Coordinates
$E=E_r$
z direction along length of the cable

Gauss's law on cylindrical surface with a radius $r > a$

$$\epsilon \oint_S \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{S}} = Q \qquad \epsilon E_r 2\pi r L = Q$$

$$E_r = \frac{Q}{2\pi\epsilon r L}$$

Voltage drop from inner to outer conductor is

$$V = -\int_b^a \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = \int_a^b E_r dr = \int_a^b \frac{Q}{2\pi\epsilon r L} dr$$

$$= \frac{Q}{2\pi L\epsilon}\int_a^b \frac{dr}{r} = \frac{Q}{2\pi L\epsilon}\ln\left(\frac{b}{a}\right)$$

Use $C = Q/V$

Capacitance of the coax segment of length L: $\boxed{C = \frac{2\pi\epsilon L}{\ln(b/a)}}$ [F]

Per unit length capacitance $\boxed{\mathcal{C} = \frac{C}{L} = \frac{2\pi\epsilon}{\ln(b/a)}}$ [F/m]

Challenge Question: Coaxial cable capacitance

- Consider a coax with ($\epsilon_{old} = 3\epsilon_0$) for $a < r < b$. If we remove the dielectric and replace it with free space ($\epsilon_{new} = \epsilon_0$) but **keep the same amount of charge** on each PC, which is **false**?

(a) the voltage across the capacitor is reduced by 3x

(b) the capacitance is reduced by 3x

(c) the E-field for a<r<b is increased by 3x

(d) the D-field for a<r<b is unchanged

Challenge Question: Coaxial cable capacitance

- Consider a coax with ($\epsilon_{old} = 3\epsilon_0$) for $a < r < b$. If we remove the dielectric and replace it with free space ($\epsilon_{new} = \epsilon_0$) but **keep the same voltage** across the coax, which is **false**:

(a) the charge across the capacitor is reduced by 3x

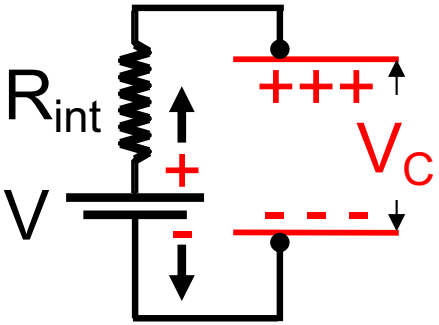
(b) the capacitance is reduced by 3x

(c) the E-field for a<r<b is reduced by 3x

(d) the D-field for a<r<b is reduced by 3x

Energy Stored in a Capacitor

- The work by the battery to move a charge dq from the bottom to the top plate is:



$$dU = V_c dq = \frac{q}{C} dq$$

V_C is instantaneous voltage on the capacitor.
It starts at zero and increases to V

- Thus the total stored energy while charging is:

$$W = \int_0^Q dq \frac{q}{C} = \frac{Q^2}{2C} = \frac{1}{2} C V_C^2$$

$$W_{\substack{\text{fully} \\ \text{charged}}} = \frac{1}{2} C V^2$$

because $V_c = V$
after charging

Stored energy of a capacitor

- Stored energy is:

$$W = \frac{1}{2}CV^2 = \frac{1}{2}\epsilon_o |E_x|^2 Ad$$ Fully charged capacitor

- Stored energy can be written is

$$W = \frac{1}{2}\epsilon_o |E_x|^2 \times Ad$$

$$\mathrm{w} = \frac{1}{2}\epsilon |E_x|^2 = \frac{1}{2}\epsilon \vec{E} \cdot \vec{E}$$

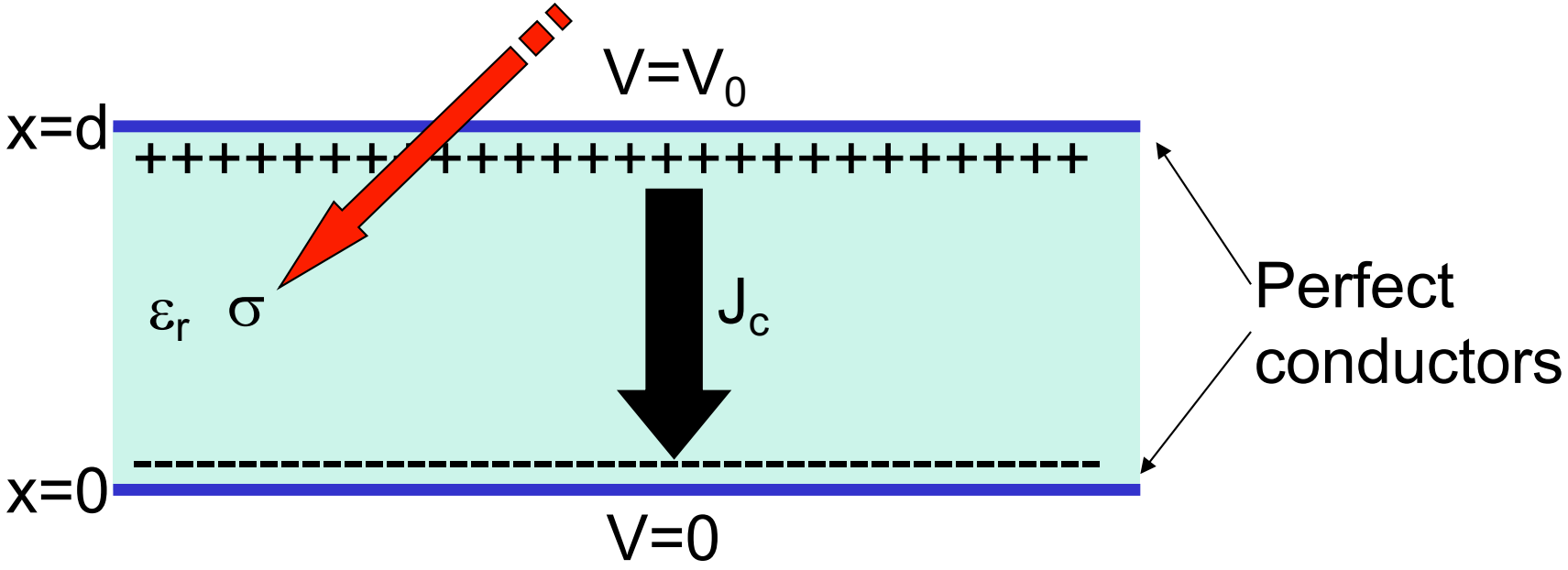
stored electrostatic energy per unit volume

$$Vol = Ad$$

volume of the capacitor

Conductance

Medium in capacitor now has conductivity



Now a current can flow x=d to x=0

$$\vec{\boldsymbol{J}} = \sigma\vec{\boldsymbol{E}} = \sigma\left(\frac{V_0}{d}\right)(-\hat{x})$$

Current density (A/m^2)

Conductance

$$G = \frac{|I_c|}{V_0} \qquad \text{or } I = GV$$

Conductance
Units: Siemens

For the parallel plate capacitor

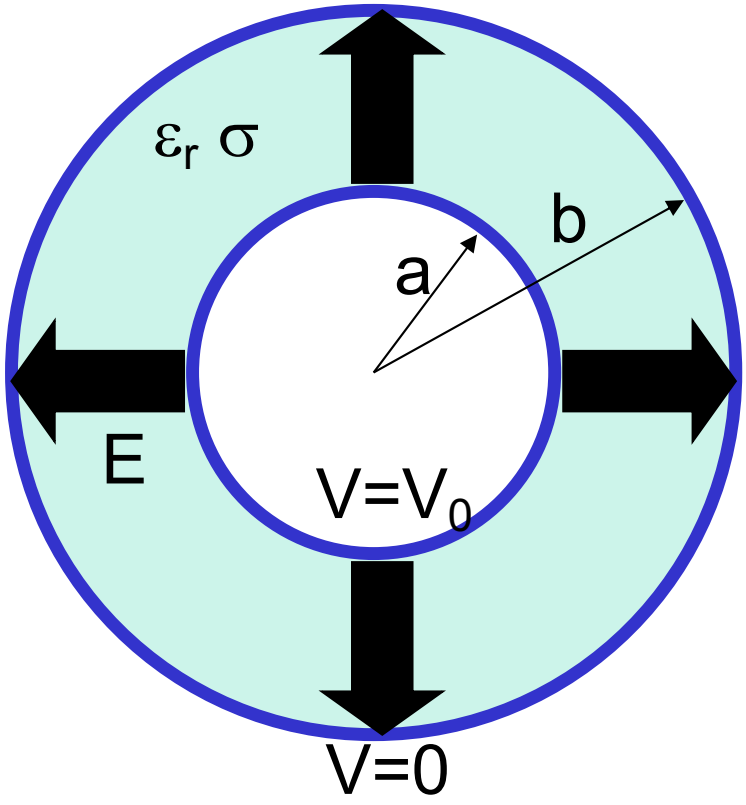
$$\vec{J} = \sigma\vec{E} = \sigma\left(\frac{V_0}{d}\right)(-\hat{x}) \qquad I = JA$$

Conductance $\boxed{G = \frac{\sigma A}{d}}$ Capacitance $\boxed{C = \frac{\varepsilon A}{d}}$

Relation between conductance and capacitance

$$\boxed{G = \frac{\sigma}{\epsilon}C}$$

Conductance/Length for Coaxial Cable

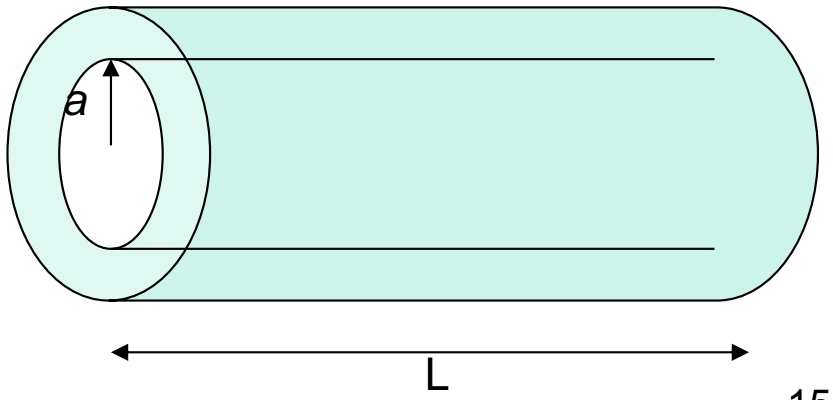


$$\vec{E} = \frac{V_0}{\ln(b/a)}\left(\frac{1}{r}\right)\vec{a}_r$$

so

$$\vec{J}_c = \sigma\vec{E} = \frac{\sigma V_0}{\ln(b/a)}\left(\frac{1}{r}\right)\vec{a}_r \qquad \left[\frac{A}{m^2}\right]$$

What is radial current (I_c) for a fixed length of the cable?



Conductance/Length of Coaxial Cable

$$I_c = \int \vec{J} \bullet d\vec{S} \qquad [A]$$

$$I_c = \int_{\phi=0}^{2\pi} \int_{z=0}^{L} J_c (r d\phi dz)$$

$$I_c = \frac{2\pi\sigma V_0 L}{\ln(b/a)}$$

$$G = \frac{I_c}{V_0} = \frac{2\pi\sigma L}{\ln(b/a)} \qquad [Siemens]$$

Conductance per unit length $\mathcal{G} = \frac{G}{L} = \frac{2\pi\sigma}{\ln(b/a)} \qquad [S/m]$

Steps for Finding Conductance

- Find Electric Field
- Find Conduction Current Density (A/m^2)

$$\vec{J}_c = \sigma \vec{E}$$

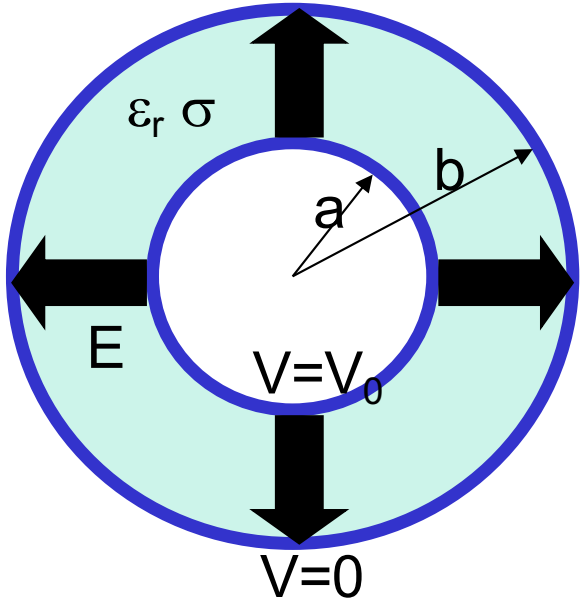
- Conduction Current (A) $I_c = \int \vec{J} \bullet d\vec{S}$
- Conductance $G = \dfrac{I_c}{V_0}$
- Conductance per unit Length

Conductance/Length for Coaxial Cable

(Review at home)

Given V, unknown Q or I

$$V \rightarrow \vec{E} \rightarrow \vec{J}$$



Step 1: Laplace's Equation

$$\nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0 \qquad a < r < b$$

$$r\frac{\partial V}{\partial r} = constant = A \rightarrow \frac{\partial V}{\partial r} = \frac{A}{r}$$

$$\rightarrow V(r) = A\ln(r) + B$$

Use boundary conditions $V(a) = V_0$ and $V(b) = 0$

$$V = A\ln(r) + B \Rightarrow V = V_0\frac{\ln(b/r)}{\ln(b/a)}$$

Step 2: $$\vec{E} = -\nabla V = -\frac{\partial V}{\partial r}\hat{r} = \frac{V_0}{\ln(b/a)}\left(\frac{1}{r}\right)\hat{r} \quad \left[\frac{V}{m}\right]$$

Step 3: $$\vec{J} = \sigma\vec{E} = \frac{\sigma V_0}{\ln(b/a)}\left(\frac{1}{r}\right)\hat{r} \quad \left[\frac{A}{m^2}\right]$$

Conductance/Length of Coaxial Cable

(Review at home)

$$V \rightarrow \vec{E} \rightarrow \vec{J} \rightarrow I \rightarrow G$$

Step 4:

$$I = \int \vec{J} \cdot d\vec{S} \qquad [A]$$

$$I = \int_{\phi=0}^{2\pi} \int_{0}^{l} \frac{\sigma V_0}{ln(b/a)} \left(\frac{1}{r}\right) (r d\phi dz)$$

$$I = \frac{2\pi\sigma V_0 l}{ln(b/a)}$$

Step 5:

conductance for cable with length L

$$G = \frac{I}{V_0} = \frac{2\pi\sigma l}{\ln(b/a)} \qquad [Siemens]$$

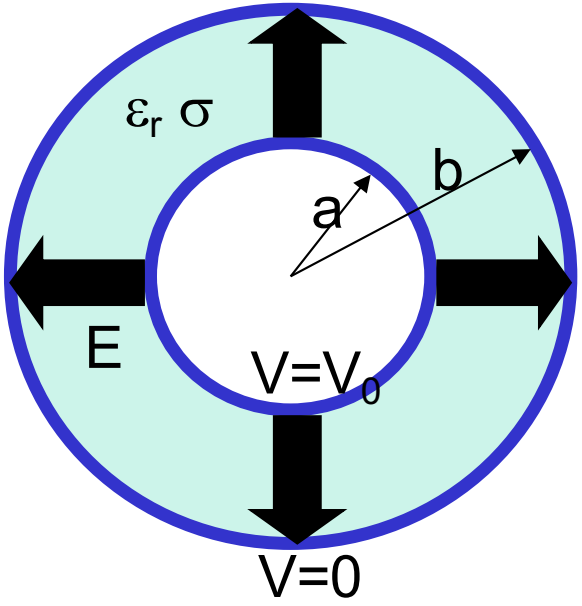
per unit length conductance

$$\mathcal{G} = \frac{G}{l} = \frac{2\pi\sigma}{\ln(b/a)} \qquad [S/m]$$

$$G = \frac{\sigma}{\epsilon} C$$

Relationship between G and C

Conductance/Length for Coaxial Cable

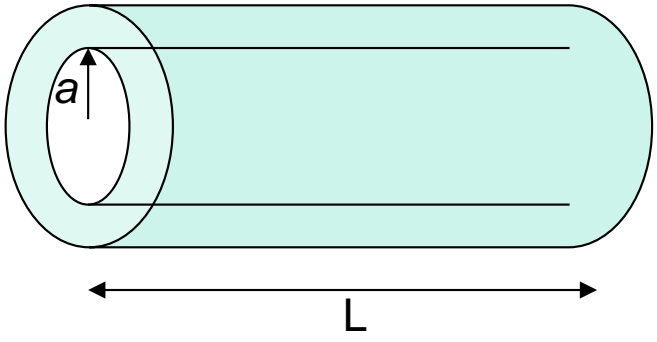


Relationship between G and C

$$G = \frac{\sigma}{\epsilon} C$$

Conductance for cable with length L

$$G = \frac{I}{V} = \frac{2\pi\sigma l}{\ln(b/a)} \quad [Siemens]$$



Per unit length conductance

$$\mathcal{G} = \frac{G}{l} = \frac{2\pi\sigma}{\ln(b/a)} \quad [S/m]$$

Lecture 10 Summary

- Capacitance $C = Q/V_0$
- Conductance $G = |I_c|/V_0$
- Next Up
  - Conductivity and Susceptibility